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# Comparison of Post-Quantum Cryptography Algorithms for Authentication in Quantum Key Distribution Classical Channel

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***Abstract*—Post-quantum cryptography is becoming more popular due to the threat posed by quantum computers, which can defeat the majority of the public key cryptography primitives currently in use, such as discrete logarithmic problems, and digital signatures. Driven by the need to protect the digital infrastructure from the possible threat of quantum computers. Continuous research, collaboration, and evaluation are required to assure the development of robust and practical post-quantum cryptography systems capable of protecting sensitive data in the post-quantum era. Quantum Key Distribution classical channel currently uses classical cryptography algorithms that are not quantum-safe. To protect the classical channel from quantum computers. Current QKD schemes do not provide authentication and data confidentiality to overcome these insecurities Post-quantum cryptography algorithms can be used. Post-quantum cryptography algorithms like Falcon (digital signature) for authentication and Crystal-Kyber (key encapsulation mechanism) for encryption in the quantum key distribution in classical channels were used.**

***Index Terms*—Quantum Key Distribution, Public key Algorithms, Quantum Cryptography**

## I. Introduction

Classical cryptography is divided into two methods symmetric cryptography and asymmetric cryptography. Asymmetric cryptography, or public-key cryptography, is used for key distribution and establishing communication between parties. Algorithms that are used for key exchange are Diffie Hellman, and Elliptical curve Diffie Hellman, and for verifying authenticity, digital signature algorithms are used. All these algorithms are based on the security of discrete logarithmic problems or integer factorization. But these problems can be solved in polynomial time using quantum computing [12]. Hence, there is a need for quantum-safe algorithms for communication, so quantum key distribution algorithms are proposed.

Quantum cryptography and quantum cryptographic techniques started in the 1980s. During this time, Bennett and Brassard introduced the groundbreaking BB84 protocol, which marked the first utilization of quantum mechanics rather than classical techniques in a cryptographic algorithm. BB84 is a Quantum Key Distribution (QKD) family protocol that uses random choices and quantum physics to establish safe key distribution between two parties. Secure key distribution is performed in many QKD methods by using two separate quantum mechanical principles: the ”No-Cloning Theorem” and the ”Heisenberg’s Uncertainty Principle” [1].

Quantum Key Distribution operates by sending many light particles, or photons, between parties over a fibre optic cable. The photons sent constitute a stream of ones and zeros, and each photon has a random quantum state. Qubits, the quantum analogue of binary bits, are the stream of quantum states that make up ones and zeros. A photon will pass through a beam splitter at the receiving end, forcing it to choose a random path into a photon collector. The sender will compare the received data from the emitter for each photon in response to the receiver data on the photon sequence. The wrong beam collectors are discarded; the left sequence is the key for encryption and decryption. Errors and data leakage are handled in the error correction and privacy amplification stages [15].

On the other hand, authentication in the classical channel is performed using asymmetric key methods that are unsecured in the quantum realm, or algorithms [2]. Inadequate security and authentication provided by these methods make the conventional channel vulnerable to security problems like man-in-the-middle attacks and eavesdropping. Consequently, both in the quantum and classical channels (where there is no intrinsic authentication), quantum-safe authentication algorithms are needed. In quantum key distribution, error correction, and further communication is performed in the classical channel. The classical channel is not a quantum-safe channel, [1] so quantum key algorithms are used for secure and quantum-safe communication.

## II. Background and Related Works

In this section, the research done on quantum key distribution is discussed.

### A. Quantum Key Distribution

Quantum key distribution (QKD) is commonly used alongside conventional symmetric encryption methods like AES and


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3DES. However, this combination is insufficient for achieving quantum safety. QKD involves transmitting photons, or quantum particles of light, over optical fibres. The first QKD protocol, invented in 1984 by Charles Bennett and Gilles Brassard, is the BB84 protocol, which is regarded as provably safe based on two conditions:

1 No-Cloning Theorem: Only by disrupting the transmission can information be gained, preventing cloning.
2 Secure Public Classic Channels: The communication of the private key among parties should occur through secure public classic channels.

The BB84 protocol is secure because the polarisation state of an individual photon cannot be monitored without destroying the particle itself. This data is still vulnerable and exposed to eavesdroppers. Any attempts to detect the photon would require the eavesdropper to reveal its presence or resend the photon, leading to errors and detection. The protocol involves Alice encoding randomly chosen bits into polarized photons (qubits) transmitted through the quantum channel. Bob, who is at the receiver end, selects a basis randomly to detect the photons generated [16].

The protocol consists of several phases:

- Key Shifting: The basis utilized to detect photons is returned to the sender by the receiver.
- Error Correction Phase: The sender returns the correct basis to the receiver, allowing both sides to eliminate bits detected using the wrong basis.
- Key Reconciliation Phase: Even after the wrong bits are discarded, the keys on the two sides may not match, necessitating the Key Reconciliation process.
- Privacy Amplification: Privacy amplification is performed to minimize information leakage.

### B. Related Works

The National Institute of Standards and Technology (NIST) has started standardizing PQC algorithms. The standards for evaluation of PQC algorithms used for security, cost, algorithm implementation features, and algorithm simplicity are taken into consideration when choosing PQC algorithms. In the third round, 15 algorithms were improved by NIST. Three of the chosen methods are digital signature or authentication schemes, while four are public-key encryption schemes. ”NTRU, Classic McEliece, SABER, and CRYSTALS-KYBER” are the public-key encryption algorithms, and ”CRYSTALSDILITHIUM, Rainbow, and FALCON” come under digital signatures used for authentication [4].

CRYSTALS-KYBER has been finalized and standardized in the Key Encapsulation category [11]. The authors of this study examine the authentication of quantum key distribution through classical channels that are not quantum-safe. As a result, the Post-quantum encryption algorithms NTRU and digital signature Falcon were used for securing the classical channel and evaluated the performance and security of the suggested technique. In this [5] white paper, the authors

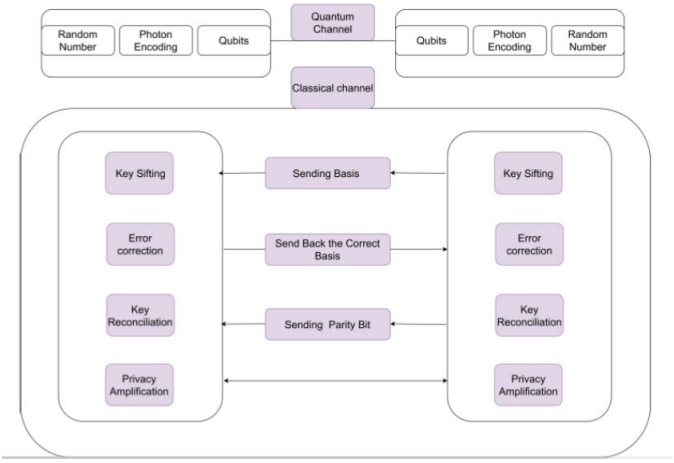


Fig. 1. Working of Quantum Key Generation System

describe quantum key distribution as a cryptographic primitive and its use cases in the real world. It was noted that QKD can be utilized in key establishment among two users, two-user Secure Payload Transmission on the basis of a Key Establishment scheme, and Key Transmission across a worldwide network with multiple users. The authors demonstrated the benefits of adding new users to the QKD network as well as experimentally proving the practicality, effectiveness, and reliability of the PQC method in QKD authentication [14]. The user uses CA to mutually agree via the PQC public-key infrastructure. The combination of QKD and PQC authentication will considerably enhance and expand the application possibilities of quantum-safe communication. Several public key algorithms may contain highly confidential data to be secured [2], using classical cryptography algorithms, which are not quantum-safe, may not be the right option. This paper [13] talks about the security of authenticated encryption algorithms in the post-quantum setting. Their security approach uses the Bellare-Namprempre paradigm to generate an authenticated encryption protocol that uses quantum-resistant symmetric-key encryption against quantum attackers.

## III. Methodology

### A. Proposed Methodology

In order to provide authenticated encryption in the conventional channel, CRYSTAL-KYBER and Falcon were chosen. After conducting a theoretical and practical analysis of NIST’s finalist authentication algorithms, it became clear that Falcon’s digital signature system has a minimal public key, private key, signing time, and verification time [8]. Therefore, Falcon has been selected to provide authentication via the traditional route. When compared to other NIST round three encryption algorithms, CRYSTAL-KYBER is the safest and most simple algorithm [6], and it outperforms NTRU. The encryption algorithm CRYSTAL-KYBER can be utilized to replace the current RSA algorithm. With a key size equal


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to AES256, CRYSTAL-KYBER can offer the same level of security as current asymmetric key cryptography standards. So, for the purpose of offering encryption via the classical channel, CRYSTAL-KYBER [6] was decided to be used.

### B. Implementation of Falcon

Gentry, Peikert, and Vaikuntanathan introduced the Falcon framework in 2008 as a lattice-based cryptographic algorithm [8]. Falcon is designed as a post-quantum signature scheme to offer secure digital signatures that are resistant to attacks from quantum computers. It relies on the challenge of computing elliptic curve isogenies. To ensure efficient signature creation and verification, it incorporates lattice structure and coding theory techniques.

Falcon employs a randomly generated private key and performs a series of mathematical operations to derive the corresponding public key. When signing a message, the algorithm generates a unique digital signature using the private key. The integrity and authenticity of the message can then be validated by verifying the signature against a pair of public keys and a private key. The security of Falcon is rooted in the computational and decisional challenges of Ring-LWE (Learning with Errors on Rings), which involve solving for the secret key given only the public key and signature.

Falcon utilizes a variant of the Fiat-Shamir transform to create signatures, eliminating the need for an interactive protocol and improving efficiency [8]. The algorithm leverages coding theory techniques, including efficient polynomial arithmetic and error-correcting codes, to achieve optimal performance without compromising security. It is a versatile signature scheme suitable for various applications like secure communication, authentication, and integrity verification. Falcon aims to provide long-term security and resilience against future cryptographic advancements.

In this implementation, Falcon is utilized during the key shifting phase of authentication in the QKD classical channel. It enables one party to sign the message while the other party verifies it, preserving authenticity based on this principle, as shown in Fig. 2. Once the verification of the signature is successful the received basis is compared with the sender basis and the mismatch basis will be discarded in the Error correction phase. If the verification is failed both parties discard their basis and communication will start again.

### C. Implementation of Crystal-Kyber

Kyber is a NIST-standardized algorithm through the Post-Quantum Cryptography (PQC) standardization process. It belongs to the CRYSTALS (Cryptographic Suite for Algebraic Lattices) family of algorithms and is classified as a quantum-safe algorithm (QSA). Kyber-1024 aims to achieve security similar to AES-256 [6]. Its security is based on the complexity of solving the learning-with-errors (LWE) problem over module lattices.

In the Crystal Kyber scheme, the public key consists of vectors, while the private key is a secret matrix. Encryption involves adding noise to the plaintext and rounding the result to the nearest lattice point. Decryption is performed by decoding the process using the private key to recover the original message. In this implementation, Crystal-Kyber is utilized in the Key Reconciliation phase. Data is encrypted using the Kyber algorithm and transmitted over the channel within the classical channel. Since there is no data leakage over the channel, the privacy amplification phase can be omitted.

### D. Measure the performance and security

There are several factors that can be used to measure the performance and security of the BB84 quantum key distribution (QKD) protocol. These factors can be divided into three main categories: security, performance, and implementation. Some specific factors that could be used to evaluate the BB84 protocol in detail include:

1. Security: Eve's Information: This is a measure of the amount of information that an attacker, known as Eve, can gain about the key by eavesdropping on the transmission. In the BB84 protocol, this information should be zero, as the principles of quantum mechanics prevent Eve from gaining any information without introducing detectable errors.
2. Error Correction Efficiency: This is a measure of the effectiveness of the error correction techniques used in the BB84 protocol. These techniques are used to correct any errors or discrepancies introduced by the transmission or measurement of the photons and should be able to correct most of these errors without introducing additional errors.
3. Privacy Amplification Efficiency: This is a measure of the effectiveness of the privacy amplification techniques used in the BB84 protocol. These techniques are used to extract a secure key from the measurement outcomes and should be able to reduce the amount of information that Eve has about the key to an arbitrarily small amount.
4. Key Error Rate: This refers to the rate at which the protocol generates an incorrect key. A low key error rate is important for ensuring the security of the transmitted information.
5. Communication Efficiency: This refers to the amount of information that can be transmitted using a given

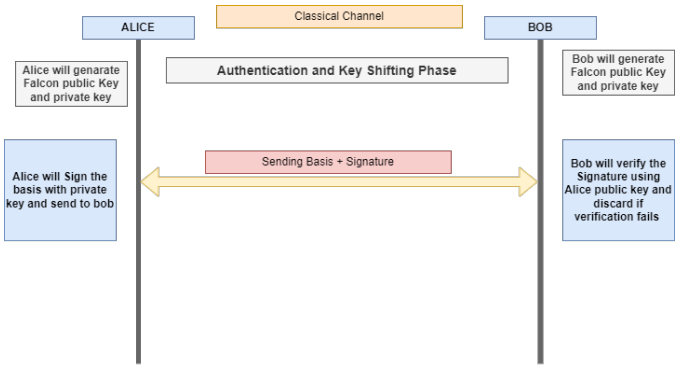


Fig. 2. Falcon in QKD Classical Channel


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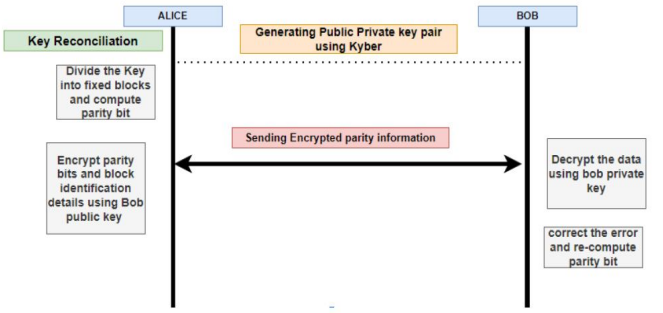


Fig. 3. CRYSTAL-KYBER in QKD Classical Channel

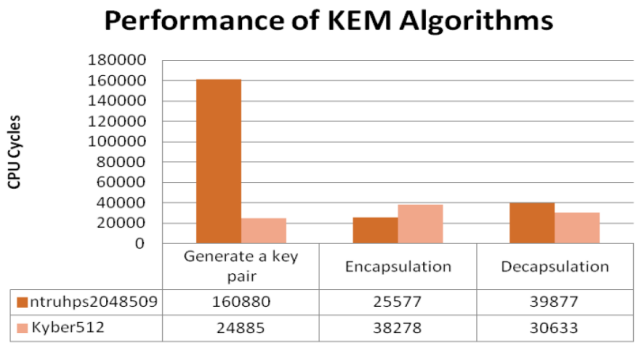


Fig. 4. Performance of NTRU versus Crystal-Kyber

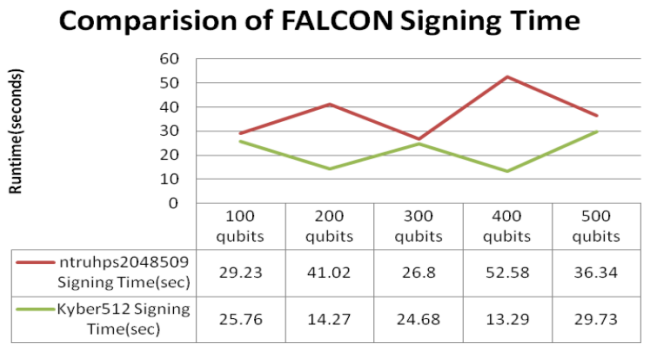


Fig. 5. Performance of Falcon Signing time NTRU versus Crystal-Kyber

amount of communication resources (e.g., bandwidth, communication channels, etc.).

6 Resistance to Attacks: This refers to the ability of the protocol to withstand various types of attacks, such as eavesdropping or tampering.

7 Robustness: The robustness of the BB84 protocol refers to its ability to withstand various types of attacks, such as noise or interference that may be present in the transmission channel. This can be measured in terms of the percentage of bits that can be successfully transmitted, or the rate at which errors are detected and corrected.

8 Distance: The distance over which the BB84 protocol can be used effectively is another important factor to consider. This can be measured in terms of the maximum distance over which the key can be transmitted without significant loss of security or efficiency.

## IV. Results and Analysis

### A. Testing Methodology

To simulate the Quantum Key Distribution (QKD) network, the powerful QuNetSim simulator was utilized. This simulator is specifically tailored for quantum networking simulations and can efficiently handle quantum and classical channel communication tasks. By seamlessly integrating Falcon and Crystal-Kyber for the QKD Classical channel in the QuNetSim simulator, a thorough test and evaluation of the performance and security aspects of the proposed scheme within a simulated quantum communication environment could be conducted [17].

### B. Performance of Crystal Kyber versus NTRU

The performance of both the algorithms NTRU [18] and Crystal Kyber [6] was analyzed. These algorithms come under the family of Key Encapsulation Mechanism algorithms. The total runtime of both algorithms is measured and Kyber has better performance when compared to NTRU, refer to fig IV-B. Hence, Crystal Kyber was used in the QKD classical channel.

### C. Falcon Signing Time and Verification Time

The Falcon has been implemented in the QKD classical channel for authentication, and the performance for the signing time was measured. The comparison of Falcon signing time has been measured between NTRU when integrated with QKD [11] and Kyber when integrated with QKD, (fig 5). The performance of Falcon verification time for NTRU versus Kyber was also measured, as shown in Fig 6. Verification time is calculated based on how much time the receiver has taken to verify the signature of different key sizes.

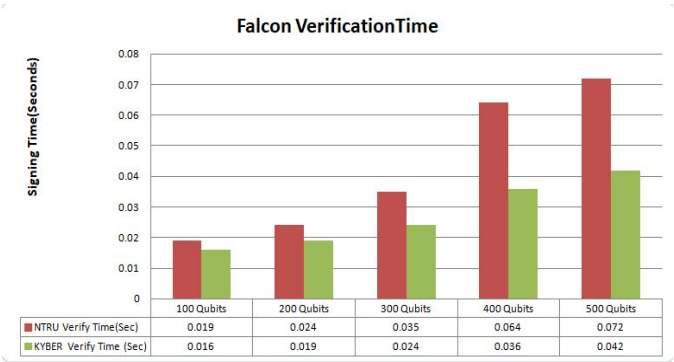


Fig. 6. Performance of Falcon Verification time NTRU versus Crystal-Kyber


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### D. Security and Performance Analysis

Several security and performance evaluations were conducted for Kyber when integrated with QKD and compared with NTRU when integrated with QKD to evaluate the overall efficiency and performance of the QKD scheme. The findings of the analysis are presented in the following sections:

*1) Correlation Coefficient Analysis:* Correlation coefficient analysis is used to measure statistical behavior and the strength of the association between two variables. A high correlation value means a strong association, while a low value suggests a weak relationship. A correlation of zero indicates no relationship. The analysis examined the correlation coefficient between the plain text and cipher text generated by the algorithm. A negative correlation value implies no correlation, indicating no relationship between the key and cipher text. Furthermore, a comparison of the correlation coefficients of Kyber and NTRU in the context of authenticated encryption was made, as shown in Table I.

TABLE I
RESULTS OF CORRELATION COEFFICIENT ANALYSIS

| Qubits Transmitted | Correlation Coefficient using NTRU | Correlation Coefficient using Kyber |
|---|---|---|
| 100 | -0.035 | -0.046 |
| 200 | -0.017 | -0.089 |
| 300 | -0.035 | -0.120 |
| 400 | -0.008 | -0.122 |
| 500 | -0.004 | -0.004 |

*2) Information Entropy Analysis:* A random variable's information entropy is an indicator of its unpredictability. In the framework of quantum key distribution (QKD), information entropy can be utilized to evaluate the security of a shared secret key. A key with a high information entropy is secure, while a key with a low information entropy is insecure. An attacker may attempt all possible bit combinations to guess the key. The number of possible permutations is 2n, where n is the key's bit count. The entropy of plain text versus encrypted text and the maximum entropy results are compared; if the encrypted entropy value is near maximum entropy or ideal entropy, it is considered a secure algorithm, as shown in table II.

*3) Key Sensitivity Analysis:* A small change in the encryption key results in a large change in the cipher text. A single bit adjustment of the encryption key was made to enable it to encrypt the qubits as part of the experiment. The encrypted text with the resulting one-bit modification is compared to the original encrypted value. The results are the similarity percentage between the two outputs. The same procedure is used for decryption as well. The experiment's outcome is expressed as a percentage. The chart clearly indicates the change in the key will cause a major change in encrypted values,as shown in table III. Comparing the results between the NTRU and Kyber. Kyber has a lower similarity percentage which means it is more secure than NTRU.

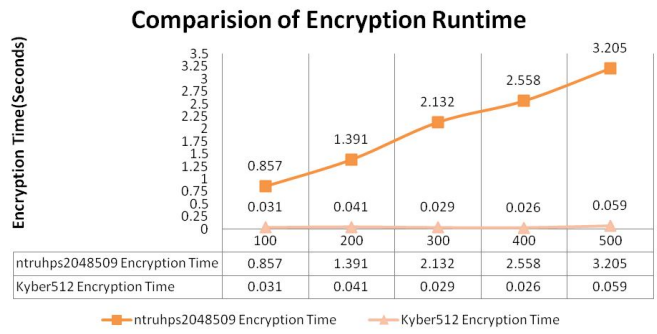


Fig. 7. Comparision of Encryption Runtime

### E. Performance Analysis

Performance analysis is the process of evaluating the efficiency and effectiveness of a QKD system. The algorithm was evaluated using the total time taken for encryption, decryption, and total execution time of the program and compared the results with the NTRU when integrated into QKD [11].

The Encryption runtime is calculated by measuring the time taken for encryption of different sizes of qubits and comparison of Kyber versus NTRU when they are integrated into QKD classical channel which is shown in Fig 7. The Decryption runtime is calculated by measuring the time taken for the decryption of different sizes of qubits and comparison of Kyber versus NTRU which are shown in Fig 8. Total runtime is calculated based on the total time taken for the algorithm and compared the total runtime of Plain QKD versus NTRU with QKD versus Kyber with QKD.

The proposed algorithm's performance (using Kyber) was evaluated by measuring the total runtime of encrypting and decrypting keys of various qubits. The runtime was compared to other techniques such as plain quantum key distribution and when NTRU is combined with QKD during the encryption stages, as shown in Fig. 7 and the decryption stages, as shown in Fig. 8. However, the results are encouraging, as the performance values of Kyber are better than those of NTRU [11], as shown in figure 9.

TABLE II
RESULTS OF INFORMATION ENTROPY ANALYSIS

| Qubits | Entropy for Plain Text | Entropy for Cipher Text | Ideal Entropy |
|---|---|---|---|
| 100 | 4.5642 | 6.4432 | 9.5849 |
| 200 | 5.2985 | 6.4424 | 9.5849 |
| 300 | 5.9991 | 6.5448 | 9.5849 |
| 400 | 5.9991 | 6.4573 | 9.5849 |
| 500 | 6.2145 | 6.4367 | 9.5849 |


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TABLE III
RESULT OF KEY SENSITIVITY ANALYSIS

| Qubits Transmitted | Key Sensitivity of Encryption using NTRU | Key Sensitivity of Encryption usingv Kyber | Key Sensitivity of Decryption using NTRU | Key Sensitivity of Decryption using Kyber |
|---|---|---|---|---|
| 100 | 0.08556 | 0.01000 | 0.00 | 0.00042 |
| 200 | 0.06221 | 0.00036 | 0.00 | 0.00063 |
| 300 | 0.03207 | 0.00012 | 0.00 | 0.00041 |
| 400 | 0.04480 | 0.00031 | 0.00 | 0.00064 |
| 500 | 0.04790 | 0.00019 | 0.415 | 0.00041 |

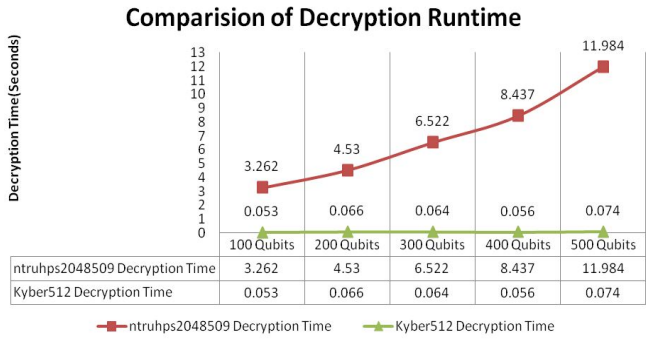


Fig. 8. Comparision of Decryption Runtime

### F. Memory Analysis

Table IV displays the memory consumption of various sizes of qubits, ranging from 100 qubits to 500 qubits, while using the proposed authentication approach. The results obtained from the comparison of Kyber when integrated into QKD versus NTRU when integrated into QKD. It shows Kyber needs less memory than other encryption techniques. This makes it an excellent solution for applications with limited memory, such as mobile devices and embedded devices.

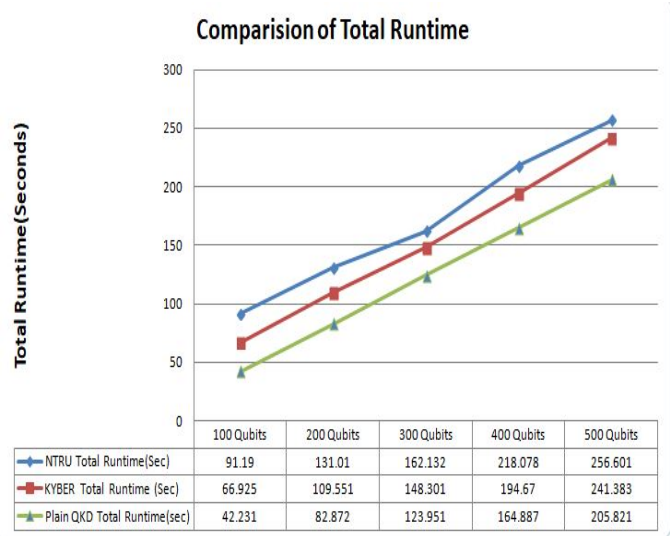


Fig. 9. Comparision of Total Runtime

TABLE IV
RESULTS OF MEMORY ANALYSIS

| Qubits Transmitted | Memory usage with NTRU (MB) | Memory usage with Kyber (MB) |
|---|---|---|
| 100 | 10.02 | 6.98 |
| 200 | 10.02 | 9.05 |
| 300 | 10.89 | 6.80 |
| 400 | 11.44 | 8.93 |
| 500 | 10.10 | 6.76 |

## V. CONCLUSION AND FUTURE WORK

Asymmetric cryptographic algorithms are currently used to secure data, but are vulnerable to attack by quantum computers. The quantum channel is inherently safe in the quantum key distribution technique, whereas the classical channel is not. Post-quantum cryptographic algorithms have been proposed to address the security issues of the classical channel. The proposed algorithm integrating Kyber into Quantum Key Distribution classical has better performance and security aspects when compared to NTRU integrated into Quantum Key Distribution classical and is more secure than existing schemes without compromising performance. Kyber is an excellent solution for applications that require security and speed compared to NTRU. As a result, Kyber can be implemented into the QKD classical channel with minimal performance loss. In the future, the proposed work can be extended to analyze the same for image encryption and measure the performance and security aspects.

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